

Supplementary Table 5. Likelihood-ratio uncertainty of the functional evidence across estimators.

Point estimates and intervals for the abnormal- and normal-class likelihood ratios under three estimands — the Haldane-corrected likelihood ratio, the Tavgigian OddsPath, and Bayesian posterior estimates — for each control set. For the likelihood ratio and OddsPath, the bounds are the log-Wald 95% confidence interval, the percentile bootstrap range, and the leave-one-out range; the Bayesian rows give the posterior median and 95% credible interval under Beta(0.5, 0.5) and Beta(1, 1) priors.

Truth set	LR type	Estimand	a	b	N path	N benign	Method	Estimate	Lo	Hi	Evidence tier	% in tier
Clinical - All variants	Abnormal	LR (Haldane)	184	25	185	1171	log-Wald	46.59	31.6	68.7	PS3 Strong	N/A (analytic)
Clinical - All variants	Abnormal	LR (Haldane)	184	25	185	1171	Bootstrap	46.59	33.3	73.2	PS3 Strong	100.0%
Clinical - All variants	Abnormal	LR (Haldane)	184	25	185	1171	LOO	46.59	46.5	48.5	PS3 Strong	100.0%
Clinical - All variants	Abnormal	OddsPath (Tavgigian)	184	25	185	1171	log-Wald	44.59	30.5	65.2	PS3 Strong	N/A (analytic)
Clinical - All variants	Abnormal	OddsPath (Tavgigian)	184	25	185	1171	Bootstrap	44.59	32	68.6	PS3 Strong	100.0%
Clinical - All variants	Abnormal	OddsPath (Tavgigian)	184	25	185	1171	LOO	44.59	44.6	46.3	PS3 Strong	100.0%
Clinical - All variants	Abnormal	Bayesian	184	25	185	1171	Beta(0.5, 0.5)	46.12	32.1	69.7	PS3 Strong	100.0%
Clinical - All variants	Abnormal	Bayesian	184	25	185	1171	Beta(1, 1)	45.13	31.4	67.7	PS3 Strong	100.0%
Clinical - All variants	Normal	LR (Haldane)	1	1133	185	1171	log-Wald	0.0056	0.000791	0.0395	BS3 Strong	N/A (analytic)
Clinical - All variants	Normal	LR (Haldane)	1	1133	185	1171	Bootstrap	0.0056	0.00276	0.0169	BS3 Strong	63.9%
Clinical - All variants	Normal	LR (Haldane)	1	1133	185	1171	LOO	0.0056	0.0028	0.00562	BS3 Strong	99.9%
Clinical - All variants	Normal	OddsPath (Tavgigian)	1	1133	185	1171	log-Wald	0.0111	0.0028	0.0441	BS3 Strong	N/A (analytic)
Clinical - All variants	Normal	OddsPath (Tavgigian)	1	1133	185	1171	Bootstrap	0.0111	0.00552	0.0224	BS3 Strong	100.0%
Clinical - All variants	Normal	OddsPath (Tavgigian)	1	1133	185	1171	LOO	0.0111	0.00559	0.0112	BS3 Strong	100.0%
Clinical - All variants	Normal	Bayesian	1	1133	185	1171	Beta(0.5, 0.5)	0.0066	0.000592	0.0256	BS3 Strong	79.4%
Clinical - All variants	Normal	Bayesian	1	1133	185	1171	Beta(1, 1)	0.0092	0.00138	0.03	BS3 Strong	90.5%
Clinical - Missense only	Abnormal	LR (Haldane)	21	0	21	28	log-Wald	56.68*	3.63	885	PS3 Strong	N/A (analytic)
Clinical - Missense only	Abnormal	LR (Haldane)	21	0	21	28	Bootstrap	56.68*	56.7	56.7	PS3 Strong	100.0%
Clinical - Missense only	Abnormal	LR (Haldane)	21	0	21	28	LOO	56.68*	54.7	56.6	PS3 Strong	100.0%
Clinical - Missense only	Abnormal	OddsPath (Tavgigian)	21	0	21	28	log-Wald	27.68	4.03	190	PS3 Strong	N/A (analytic)
Clinical - Missense only	Abnormal	OddsPath (Tavgigian)	21	0	21	28	Bootstrap	27.68	27.7	27.7	PS3 Strong	100.0%
Clinical - Missense only	Abnormal	OddsPath (Tavgigian)	21	0	21	28	LOO	27.68	26.7	27.6	PS3 Strong	100.0%
Clinical - Missense only	Abnormal	Bayesian	21	0	21	28	Beta(0.5, 0.5)	123	11.8	5.84e+04	PS3 Strong	61.3%
Clinical - Missense only	Abnormal	Bayesian	21	0	21	28	Beta(1, 1)	40.33	8.1	1.12e+03	PS3 Strong	71.0%
Clinical - Missense only	Normal	LR (Haldane)	0	28	21	28	log-Wald	0.0231*	0.00149	0.358	BS3 Strong	N/A (analytic)
Clinical - Missense only	Normal	LR (Haldane)	0	28	21	28	Bootstrap	0.0231*	0.0231	0.0231	BS3 Strong	100.0%
Clinical - Missense only	Normal	LR (Haldane)	0	28	21	28	LOO	0.0231*	0.0231	0.0242	BS3 Strong	100.0%
Clinical - Missense only	Normal	OddsPath (Tavgigian)	0	28	21	28	log-Wald	0.0471	0.00693	0.32	BS3 Strong	N/A (analytic)
Clinical - Missense only	Normal	OddsPath (Tavgigian)	0	28	21	28	Bootstrap	0.0471	0.0471	0.0471	BS3 Strong	100.0%
Clinical - Missense only	Normal	OddsPath (Tavgigian)	0	28	21	28	LOO	0.0471	0.0471	0.0493	BS3 Strong	100.0%
Clinical - Missense only	Normal	Bayesian	0	28	21	28	Beta(0.5, 0.5)	0.0109	2.6e-05	0.116	BS3 Strong	60.0%

Truth set	LR type	Estimand	a	b	N path	N benign	Method	Estimate	Lo	Hi	Evidence tier	% in tier
Clinical - Missense only	Normal	Bayesian	0	28	21	28	Beta(1, 1)	0.0320	0.00124	0.161	BS3 Strong	62.7%
KURIS/NDD vs B/LB missense	Abnormal	LR (Haldane)	14	0	17	28	log-Wald	46.72*	2.97	736	PS3 Strong	N/A (analytic)
KURIS/NDD vs B/LB missense	Abnormal	LR (Haldane)	14	0	17	28	Bootstrap	46.72*	37.1	56.4	PS3 Strong	100.0%
KURIS/NDD vs B/LB missense	Abnormal	LR (Haldane)	14	0	17	28	LOO	46.72*	45.1	49.5	PS3 Strong	100.0%
KURIS/NDD vs B/LB missense	Abnormal	OddsPath (Tavtigian)	14	0	17	28	log-Wald	22.56	3.24	157	PS3 Strong	N/A (analytic)
KURIS/NDD vs B/LB missense	Abnormal	OddsPath (Tavtigian)	14	0	17	28	Bootstrap	22.56	17.7	27.4	PS3 Strong	93.4%
KURIS/NDD vs B/LB missense	Abnormal	OddsPath (Tavtigian)	14	0	17	28	LOO	22.56	21.8	23.9	PS3 Strong	100.0%
KURIS/NDD vs B/LB missense	Abnormal	Bayesian	14	0	17	28	Beta(0.5, 0.5)	101	9.13	4.32e+04	PS3 Strong	59.9%
KURIS/NDD vs B/LB missense	Abnormal	Bayesian	14	0	17	28	Beta(1, 1)	33.83	6.44	841	PS3 Strong	65.6%
KURIS/NDD vs B/LB missense	Normal	LR (Haldane)	3	28	17	28	log-Wald	0.1765	0.0632	0.493	BS3 Moderate	N/A (analytic)
KURIS/NDD vs B/LB missense	Normal	LR (Haldane)	3	28	17	28	Bootstrap	0.1765	0.0283	0.353	BS3 Moderate	61.6%
KURIS/NDD vs B/LB missense	Normal	LR (Haldane)	3	28	17	28	LOO	0.1765	0.125	0.188	BS3 Moderate	100.0%
KURIS/NDD vs B/LB missense	Normal	OddsPath (Tavtigian)	3	28	17	28	log-Wald	0.2302	0.0967	0.548	BS3 Moderate	N/A (analytic)
KURIS/NDD vs B/LB missense	Normal	OddsPath (Tavtigian)	3	28	17	28	Bootstrap	0.2302	0.0575	0.403	BS3 Moderate	64.2%
KURIS/NDD vs B/LB missense	Normal	OddsPath (Tavtigian)	3	28	17	28	LOO	0.2302	0.183	0.244	BS3 Moderate	68.9%
KURIS/NDD vs B/LB missense	Normal	Bayesian	3	28	17	28	Beta(0.5, 0.5)	0.1869	0.0533	0.413	BS3 Moderate	64.6%
KURIS/NDD vs B/LB missense	Normal	Bayesian	3	28	17	28	Beta(1, 1)	0.2065	0.0662	0.43	BS3 Moderate	58.7%

Estimands: LR (Haldane) = $(a/N_{\text{path}})/(b/N_{\text{ben}})$; OddsPath (Tavtigian) adds +1 to whichever in-class count is the error cell for that row -- benign $(a/(N_{\text{path}}+1))/(b/(N_{\text{ben}}+1))$ for Abnormal rows, pathogenic $((a+1)/(N_{\text{path}}+1))/(b/(N_{\text{ben}}+1))$ for Normal rows; Bayesian = posterior-median LR under Beta(0.5,0.5) (Jeffreys) and Beta(1,1) (uniform) priors. The LR (Haldane) point and its log-Wald CI are the primary baseline; the others are stability checks.

Lo / Hi: analytic 95% CI (log-Wald); 95% percentile interval (Bootstrap); min and max across refits (LOO = leave-one-out, each control dropped once); 95% credible interval (Bayesian). '% in tier' is the fraction of resamples/draws in the estimand's full-data ACMG tier.

* Haldane-Anscombe correction (+0.5 to all four cells) applied when a functional-class cell is empty.